Vania Janet Raya Ríos

Mexico City, Mexico | vani.janet.cd@gmail.com | LinkedIn | GitHub | ORCID
Personal website: [URL TO ADD] | Google Scholar: [URL TO ADD, IF AVAILABLE]

Education

National Autonomous University of Mexico (UNAM), IIMAS 2022–2026

B.Sc. in Data Science; coursework completed; degree award / professional examination pending confirmation. **Reported GPA: 9.55/10**.

Undergraduate thesis: *Recuperación híbrida multi-turno en dominios especializados: ablación de fusión, reescritura y reranking con implicaciones para producción.* English translation: "Hybrid Multi-Turn Retrieval in Specialized Domains: Ablation of Fusion, Rewriting, and Reranking with Implications for Production."

Relevant coursework: machine learning; probability and stochastic simulation; optimization; linear algebra; numerical methods; concurrent/distributed computing; databases.

Peer-Reviewed Publications

1. **V. Raya-Rios**, H. Gómez-Adorno, L. Hecht, P. Vázquez-Osorio, E. Fabián-Sandoval, J. Vázquez-Osorio, and D. Hernández-Bustamante. "IIMAS-RAG at SemEval-2026 Task 8: Hybrid Sparse-Dense Retrieval and Answerability-Conditioned Generation for Multi-Turn RAG." In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval 2026)*, pp. 2744–2753, 2026. [DOI / paper] [Code]. **Best System Paper Award**; retrieval **4th / 38** (nDCG@5 0.5445).

2. **V. Raya-Rios***, A. Klimchuk*, N. M. Cuadrado, I. Gollini Navarrete, S. Horvath, and M. Takac. "Tiny Titans at WMT26: Task-Calibrated Quantization and Activation-Aware Low-Rank Compression of Gemma 3 12B." *Proceedings of the Eleventh Conference on Machine Translation (WMT 2026)*, pp. 2037–2044, scheduled October 28–29, 2026. [Proceedings PDF]. *Equal contribution. Accepted; conference forthcoming as of October 18, 2026.

Research Experience

Visiting Researcher, University of Zurich (UZH), Switzerland Jul–Aug 2026

Supervisor: Dr. Selene Báez Santamaría. Studied phenotype extraction and Human Phenotype Ontology (HPO) normalization for rare-disease information retrieval; implemented and evaluated neural mention extraction, candidate retrieval and LLM verification; investigated hierarchical selective prediction and risk control.

Undergraduate Research Intern, MBZUAI, United Arab Emirates (UGRIP) Jun–Jul 2026

Supervisors: Martin Takac and Samuel Horvath; mentor: Nicolas M. Cuadrado. **Equal-first-author WMT26 collaboration**. Developed task-calibrated 4-bit GPTQ workstream for Czech-to-German translation with Gemma 3 12B. On a 300-sentence development set, **99.7%** of BF16 COMET score retained with GPU memory decreasing **26.6 to 17.6 GiB**. Team compared activation-aware SVD (**46.6%** parameter reduction) and combined compression approaches.

NLP Researcher, IIMAS–UNAM, Mexico 2025–2026

Supervisor: Dr. Helena Gómez-Adorno. **First author** of SemEval 2026 team system paper. Investigated multi-turn query rewriting, SPLADE + Voyage dense–sparse fusion via RRF, and answerability-conditioned generation. LLM query rewriting improved nDCG@10 by **16.3% relative** over the tested no-rewrite hybrid baseline; full RAG ranked 13th/29.

Awards and Honours

- **Best System Paper Award**, SemEval 2026 Task 8 (IIMAS-RAG). [UNAM recognition]. [Official individual award certificate: TO ADD]
- **First Place**, Samsung México–IIMAS–UNAM, *Miradas al Mañana* essay competition, June 2026. Essay: "Ver con las manos, oír con los ojos: IA accesible en pantallas para inclusión y sostenibilidad" ("Seeing with Our Hands, Hearing with Our Eyes: Accessible AI in Displays for Inclusion and Sustainability"). [Official announcement] [Spanish essay] [English translation]. [Certificate: TO ADD]
- **MBZUAI UGRIP 2026 selection**; institution reported acceptance rate of **2.2%**. [UNAM announcement]. [Selection letter: TO ADD]
- **AI Buildathon Winner**, Product-LatAm Summit / Mexico Tech Week 2025 (team AI Sales Copilot). [Project]. [Official winner certificate: TO ADD]

Industry Experience

Associate Consultant Intern, Performance Analytics – Mastercard Dec 2025–Jun 2026

Automated data extraction, quality controls, segmentation and reporting in Python, SQL and Impala; reported reducing a recurring analytics cycle from 8–13 to 1–2.5 hours (~**80%** reduction).

Data Science Intern – PagaTodo (FinTech) Jul–Aug 2025

Designed and evaluated agentic/RAG workflows connected to internal information and business tools; optimized vector retrieval for a support pipeline (reported **80% latency reduction**).

Freelance Chatbot Developer – HSK Insurance Solutions, Colombia Jun 2024–Jul 2025

Designed insurance customer-service chatbot flows, analytics, and Meta WhatsApp/Facebook messaging integrations.

Freelance AI Delivery Chatbot Developer – MotoYa, Colombia Dec 2023–May 2024

Built Python/Twilio WhatsApp order-handling and driver-dispatch workflows on AWS EC2, including voice and sticker inputs.

Teaching, Outreach and Selected Projects

- **Teaching Assistant**, Paradigmas y Programación, UNAM (2026): student instruction and academic mentoring. **ACL 2026**: SemEval research poster presenter and student volunteer (San Diego).
- **Soy Mujer y Hago Ciencia**: volunteer graph-theory workshop (Eulerian paths and circuits); promoted participation in science among younger students.
- Multi-turn RAG retrieval benchmarks (evaluation, ablations, Docker); Seq2Seq reproducibility (PyTorch, controlled ablations); LLM essay reviewer (rubric-based scoring, human oversight).

Skills and Competencies

Programming / data: Python, SQL, R, C++; pandas, NumPy, Spark, Hadoop, Hive. **Machine learning / NLP:** PyTorch, scikit-learn, Hugging Face, neural retrieval, SPLADE, RRF, RAG, quantization (GPTQ), knowledge distillation, evaluation/ablations. **Infrastructure:** Linux, Docker, Slurm/GPU research workflows, AWS, Azure. **Languages:** Spanish (native); English (professional working proficiency; test evidence: TO ADD).